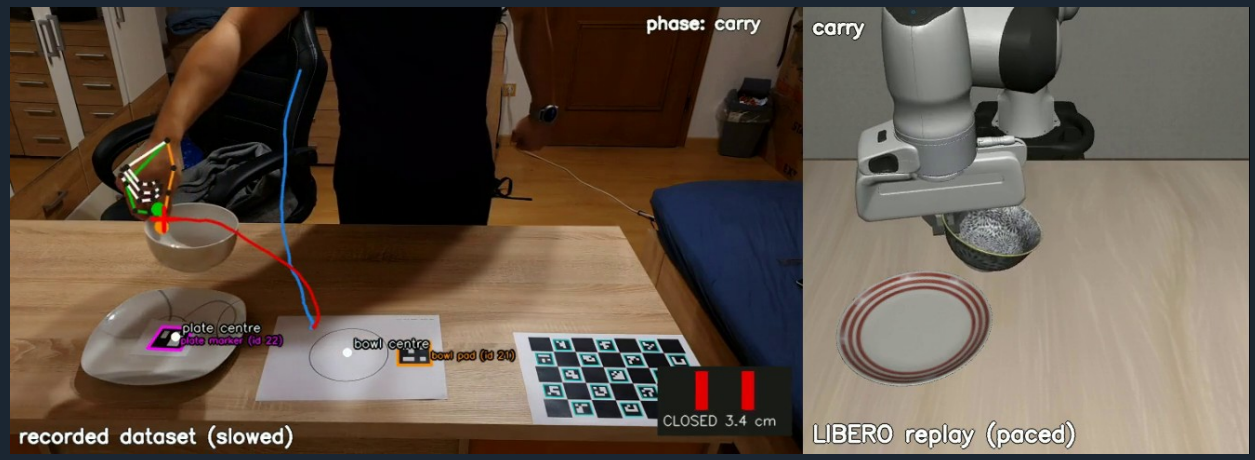
hurobo

From phone video of a human hand to a robot policy in simulation

github.com/alfredowh/hurobo

Real → Sim



The question: can human video help?

QUESTION

Can human videos improve a small VLA on a new layout and a new object?

Test case: SmolVLA (450M), bowl onto plate in LIBERO, trained only on data from a phone video of a hand. Simulation only.

1. **One 16 GB GPU**
   450M model in bf16, no 3B/7B models.
2. **Phone only**
   ~15 tripod clips. One camera, no depth sensor, no glove.
3. **8 – 16 October 2026**
   Start of the project to the deadline.
4. **No physical robot**
   Success rate in LIBERO is the only metric.

Pipeline: hand video in, robot demos out

REAL WORLD

1 Record

Tripod phone, ChArUco-calibrated. ArUco markers on table, bowl, plate.

2 Extract

Hand TCP (thumb/index midpoint), grasp from pinch, object poses, in the table frame.

3 Twin

Copy real layout and camera into LIBERO. Real vs. sim side-by-side replay.

SIMULATION (LIBERO)

4 Multiply

Split demo into object-relative segments, retarget to many sim poses, keep only successes.

5 Train

Fine-tune SmolVLA from smolvla_base (never saw LIBERO) on generated data only.

6 Evaluate

lerobot-eval, N = 50, success rate with 95% CI against baselines.

Four ways to close the real-to-sim gap

1 Teleoperate in sim

IDEA

Control the Panda by tracked hand or keyboard in the simulator.

GAP LEFT

No visual gap.

CATCH

Jerky, slow, few demos. Real recording unused.

Debug tool only

2 Scan the objects

IDEA

Bowl and plate from a phone orbit: video, COLMAP, Gaussian splat, collision mesh.

GAP LEFT

Objects match; scene stays synthetic.

CATCH

Scan quality, metric alignment.

Stretch

3 Scan the whole scene

IDEA

Splat the table; render sim from the real view; composite robot by depth.

GAP LEFT

Smallest visual gap.

CATCH

Alignment, GPU time. Physics still LIBERO.

Stretch

4 Size-matched objects

IDEA

Use real objects with the LIBERO bowl and plate dimensions; sim assets are scaled to match.

GAP LEFT

Size matches; look does not, but the policy sees no real pixels.

CATCH

Measure objects, print markers.

Core path

Main concept and current state

Core

Motion-only retargeting

Video as motion capture: a few object-relative numbers cross to the sim.
Now: built. 12 of 26 recorded clips cleaned and usable.

Main visual

SceneTwin: real layout in sim

Real clip beside the sim replay from the same viewpoint.
Now: built. Recorded episode and sim replay side by side on the next slide.

Stretch

Gaussian-splat objects and scene

Shows visual fidelity, not a trained result.

Now: bowl, plate and scene scanned; bowl and plate shown in LIBERO.

WHO CONTRIBUTES WHAT

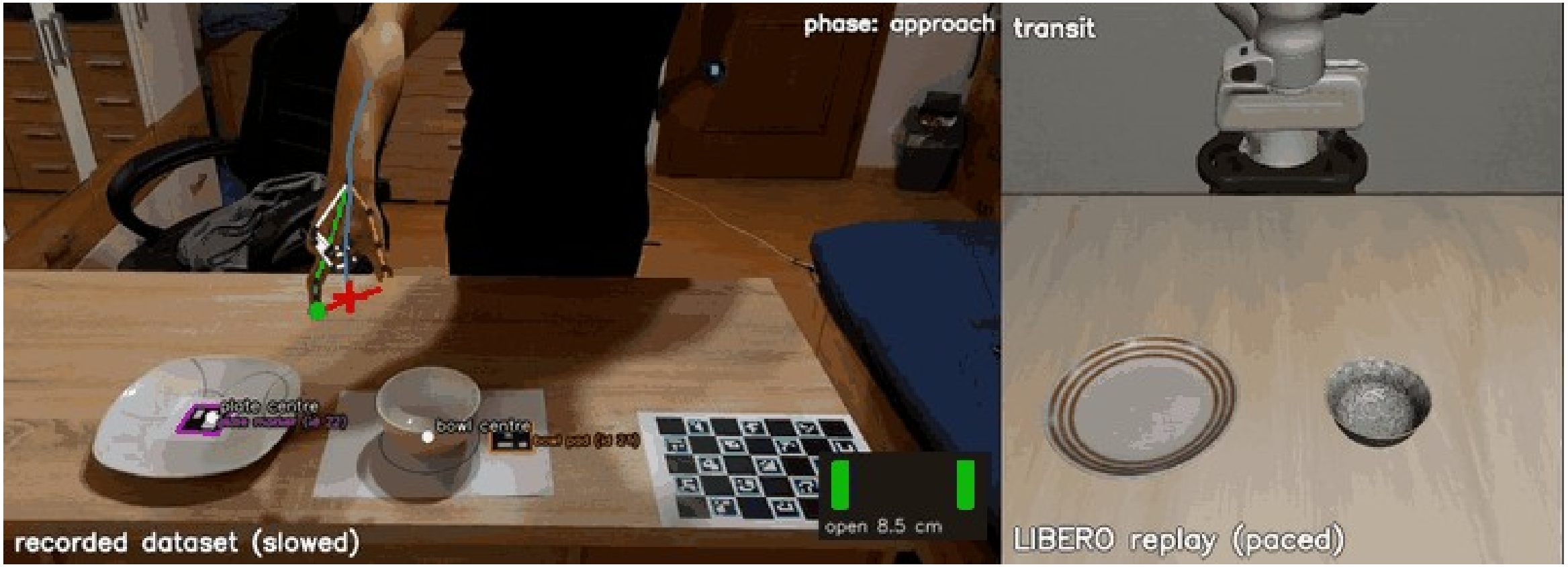
Human video

Approach direction, grasp timing, lift height, carry path, place offset, release timing.

Simulator

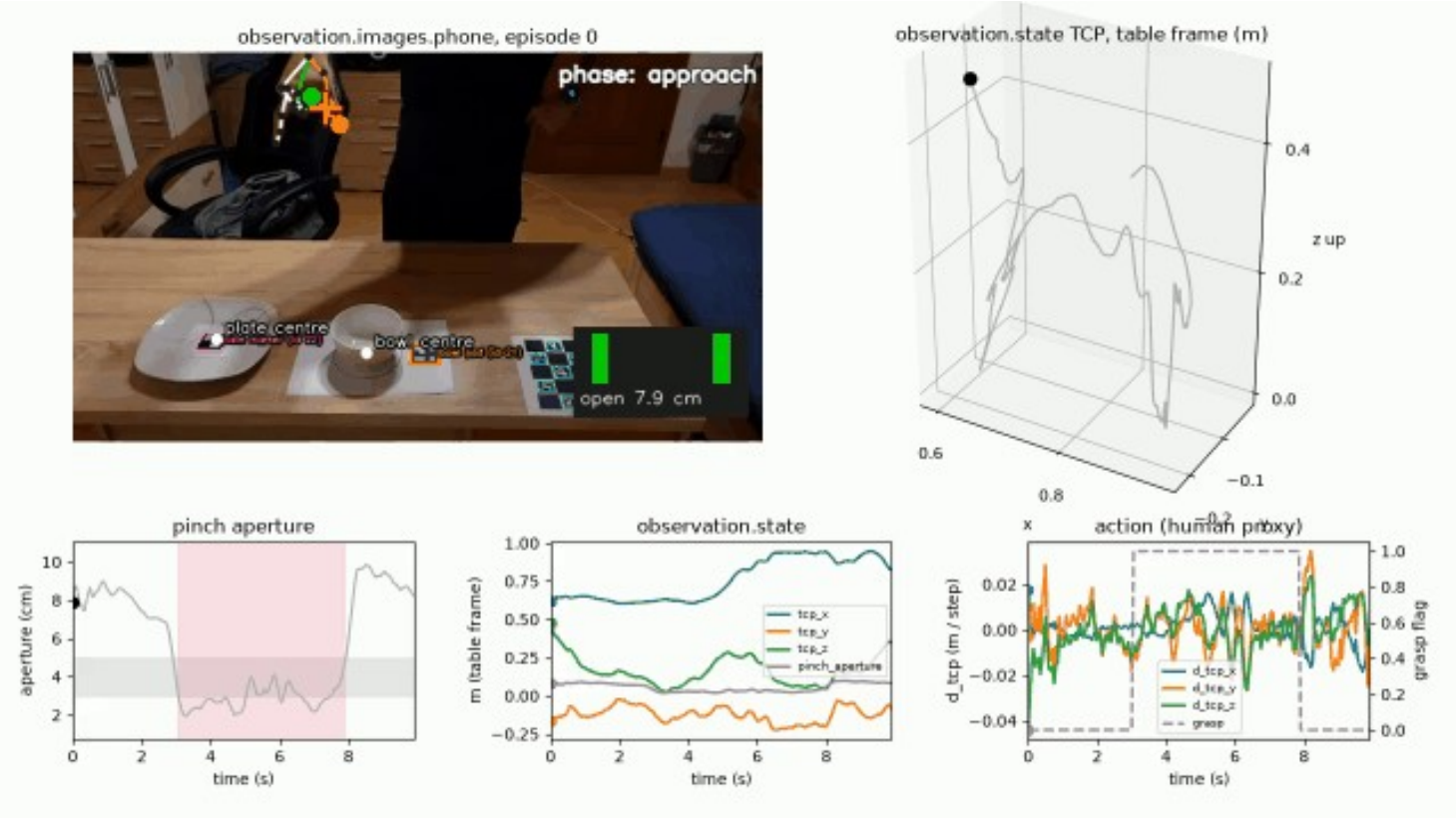
Exact object poses, geometry and appearance. Fixed top-down gripper orientation, yaw from the object.

SceneTwin: recorded clip and LIBERO replay



Recorded episode (left, slowed) beside its LIBERO replay (right, paced).

The human demo: hand and markers



Markers

ArUco on the plate (id 22) and the bowl pad (id 21); the ChArUco board is the table frame.

Hand

MediaPipe skeleton. The TCP is the midpoint of thumb and index tip, in metres in the table frame.

Gripper

Pinch aperture with hysteresis gives open or closed (here: closed).

Episode

199 samples at 20 Hz over 9.9 s, stored as a LeRobot episode.

One real clip, cut to a single grasp cycle. 12 of the 26 recorded clips were cleaned and usable.

From 26 clips to 12 usable demos

1 Filter frames

Drop frames with no hand or two hands, a marker pose that reprojects badly (over 15 px), a hand at the image border, or depth outside 0.15 to 3 m.

2 Find the cycle

Keep the first gap-free run (gaps up to 0.4 s bridged) that holds open, closed, open. Pinches under 0.2 s and open gaps under 0.3 s count as noise.

3 Cut

Trim from the start of the run to 2 s after the release, so the hand’s retreat is kept.

4 Reject

Drop clips shorter than 1.5 s, or without the board, bowl and plate marker in the first second.

Result: 12 of the 26 recorded clips are cleaned and usable, each one grasp cycle.

Generating the robot demos

1 Sample layout

Draw a random bowl and plate position in the size-matched LIBERO scene.

2 Pick sources

Draw one of the 12 cleaned human demos at random for each rollout.

3 Re-anchor

Approach: the last 20 cm of the hand path, relative to the bowl. Carry: the bowl’s path, relative to the plate, ending over its centre. Both move with the new layout.

4 Execute, keep wins

Grasp at a fixed 4 cm height, fingers facing the rim; release 4 mm over the plate. Run closed loop in LIBERO and keep only the successes.

First run, 12 clips: 100 successes of 170 attempts (59 %) in about 40 to 50 min, one process. Failures: 37 bowl not lifted, 33 timeouts.

Generated demos: camera 1 and 2

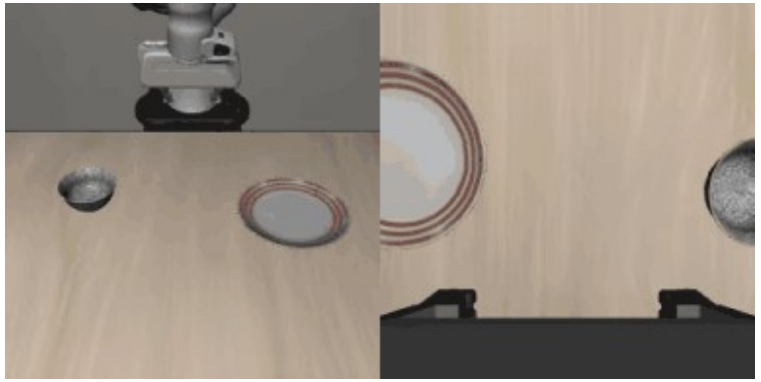
Generated episode 1



Generated episode 2

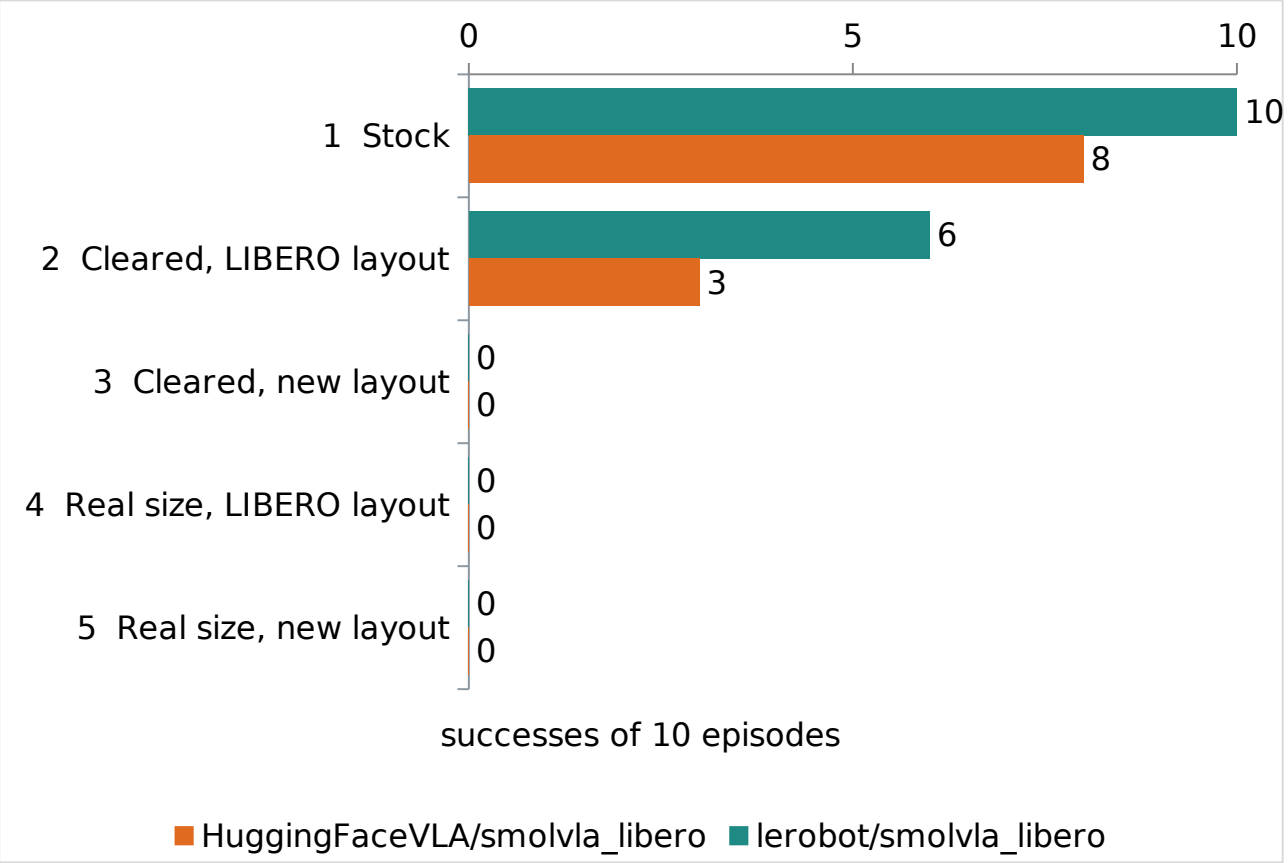


Generated episode 3



Objects are scaled to real size (bowl x1.23, plate x1.85 of LIBERO’s). Per clip: camera 1 (agent view) left, camera 2 (wrist) right.

Pretrained SmolVLA: layout and size matter



1 Stock

LIBERO's own scene and start positions.

2 Cleared, LIBERO layout

Other objects removed; bowl and plate start where LIBERO puts them.

3 Cleared, new layout

Other objects removed; bowl and plate at random positions.

4 Real size, LIBERO layout

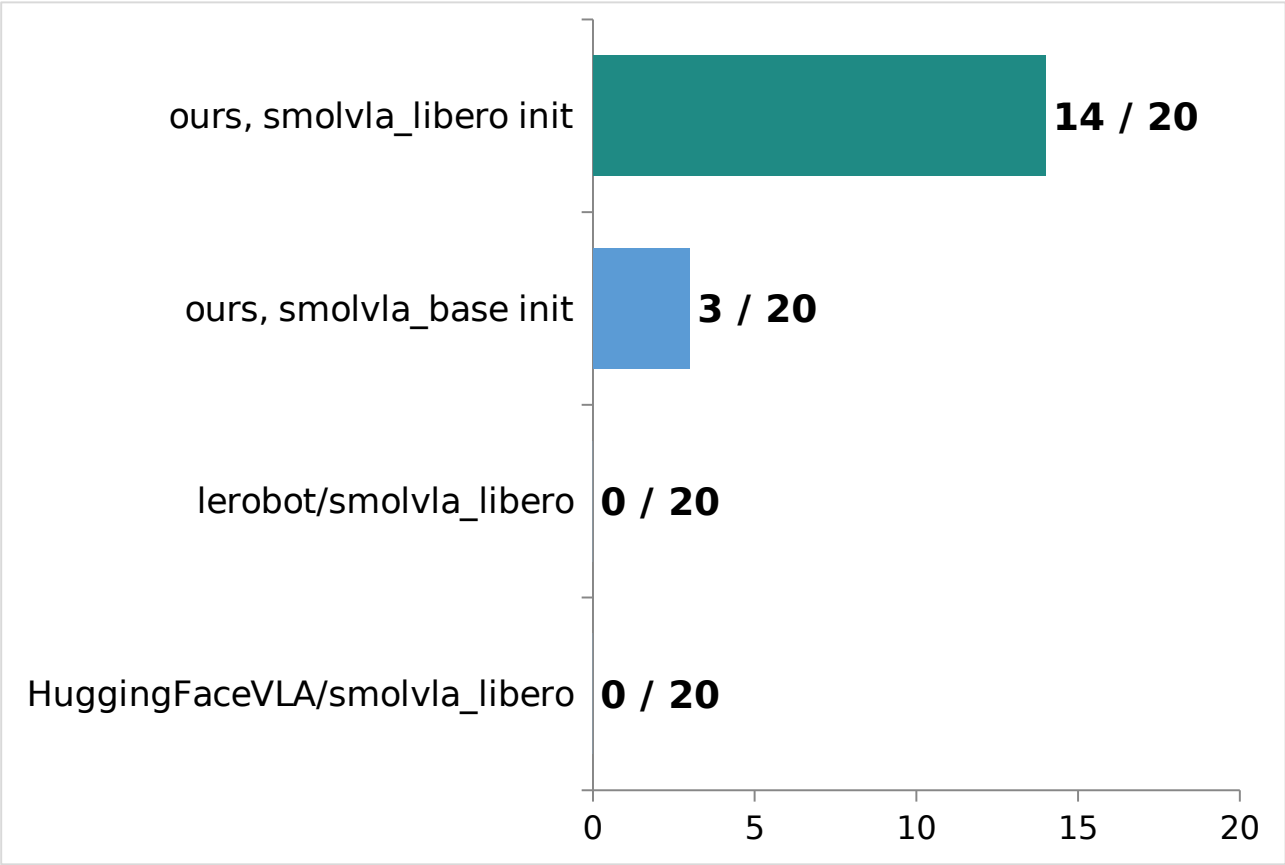
Bowl and plate at measured real sizes (plate 1.86x wider), LIBERO's placement.

5 Real size, new layout

Real sizes at random positions: the target scene of our experiment.

A new layout alone takes the better checkpoint from 6/10 to 0/10; real-size objects give 0 even on LIBERO's layout. Pretrained weights, 10 episodes per cell: 0/10 means 0 to 28% at 95%.

Our fine-tuned SmolVLA on held-out layouts



ours, smolvla_libero init

70% (95% CI 48 to 85%). Open loop: position error 0.075 against 0.137 for an all-zero chunk.

ours, smolvla_base init

15% (95% CI 5 to 36%)

lerobot/smolvla_libero

0% (95% CI 0 to 16%). Pretrained, never saw our scene.

HuggingFaceVLA/smolvla_libero

0% (95% CI 0 to 16%). Open loop: gripper sign right only 41% of the time.

Scaled scene, 20 held-out layouts, same for every policy. Starting from smolvla_libero beats starting from smolvla_base (14/20 vs 3/20, intervals do not overlap). One seed; the best checkpoint of each run was picked on other layouts (0 to 9).

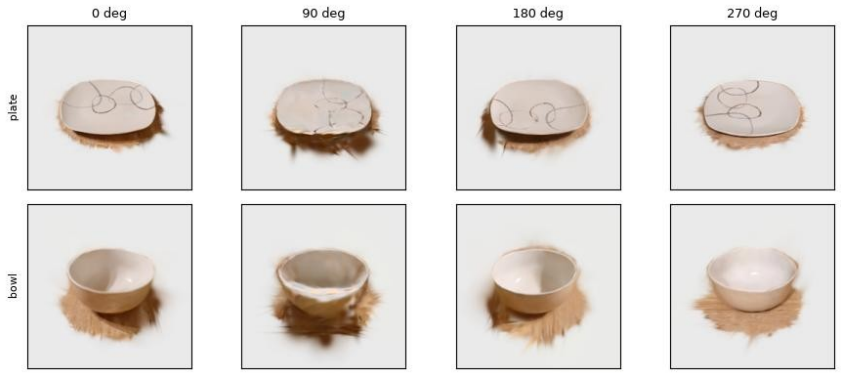
Future work: photoreal real2sim



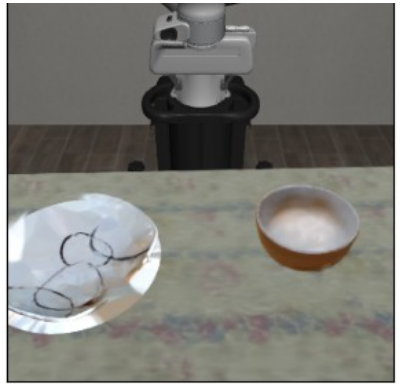
Whole scene scan: table, background, bowl, plate. Four held-out views, photo left, splat right



Whole scene in 3D (rendered splat)



Bowl and plate splats cut out of the scan, four angles.



Scanned bowl and plate inside LIBERO. The table cloth is still LIBERO's.

NEXT

Place the whole-scene splat in LIBERO

Collision meshes for bowl and plate

Composite the mesh robot by depth

Show as visual fidelity, not a trained result

Future work: faster demo generation

Result (7 clips)

Default: 38 to 47 accepted per 100 to 120 attempts

Only 15 to 21 % within 300 steps

Screened MimicGen-style, best run:

100 of 125 (80 %), all within 300 steps

Pros

About 4x more usable episodes per attempt

Every episode fits the 300-step evaluation

Drops the 3 of 7 clips that never succeed

Noise and source mixing add variety

Cons

Gain is mostly screening, speed, length filter

Control run missing

No policy trained on it yet

7 clips, one session, one seed

Optimization potential, not a result: not used for the 100-episode training set. Notebook 12, D-013.

Limitations and status

Limitations

- Simulation only, no real robot
- Hand vs. gripper, real vs. rendered pixels
- One camera, markers required
- Fixed top-down grasp, one task
- 12 usable clips; marker size not measured

Done, on real clips

- 26 clips recorded, 12 cleaned
- 100 generated episodes (100 of 170 rollouts)
- SmolVLA fine-tuned: 14/20 from smolvla_libero, 3/20 from smolvla_base; pretrained 0/20
- Bowl, plate and scene scanned as splats

Still open

- N = 50 evaluation, more seeds
- Zero-shot and open-loop replay baselines
- ACT baseline
- Scene splat and collision meshes in LIBERO
- Train on the faster generation

Every number in this deck comes from a run in the repo. Failed runs are included. One seed and N = 20: the intervals are wide.